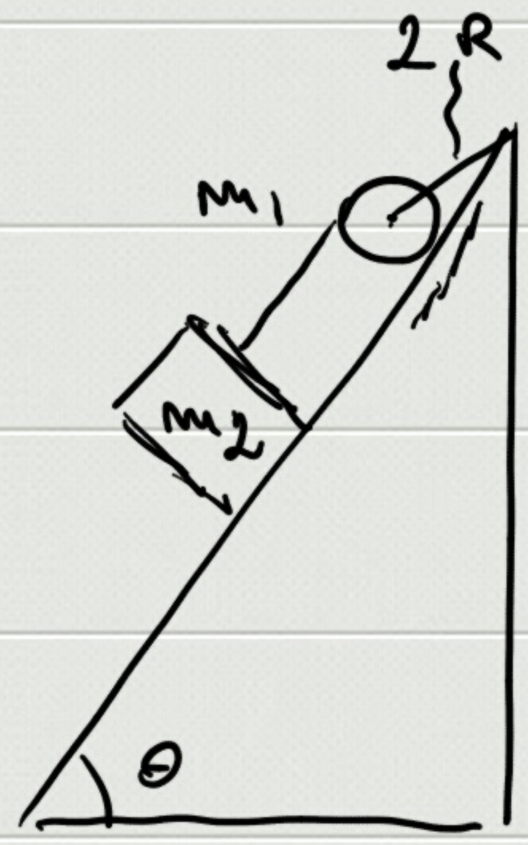


$\theta = 60°$

$m_1 = 0.5$ kg

$m_2 = 0.5$ kg

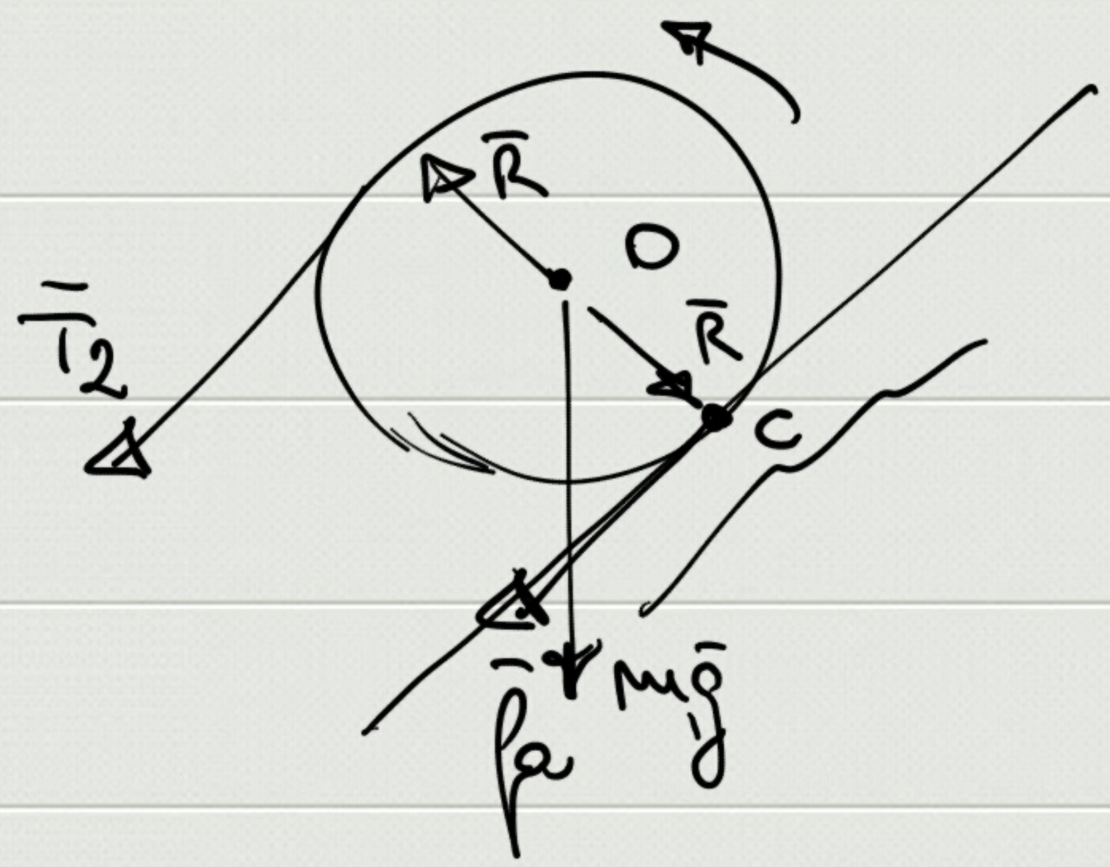
$f_a = 2$ N

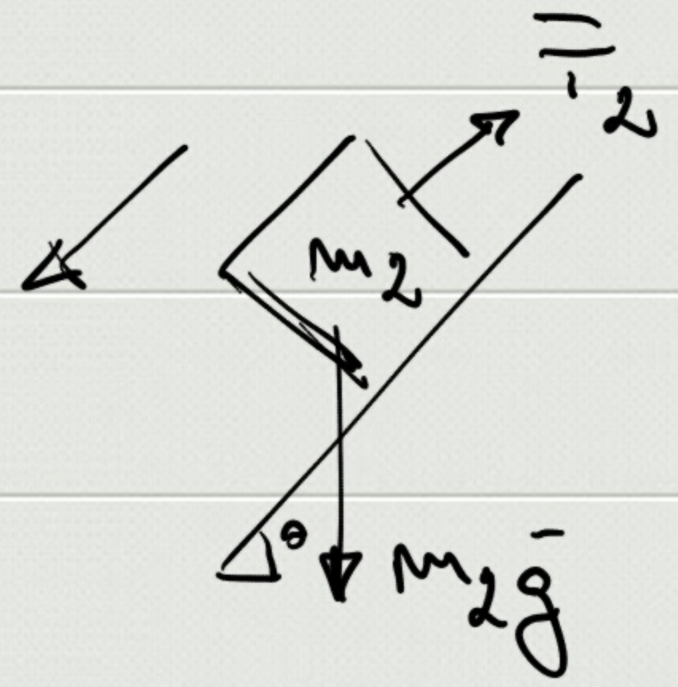
a_2 , T_2

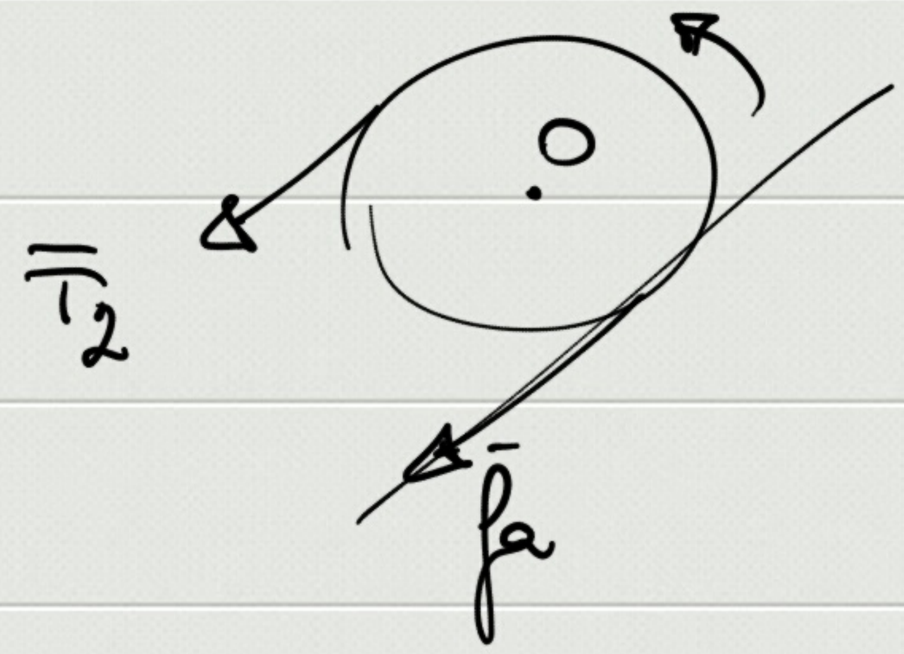
$$* \quad \boxed{R T_2 - R f_a = \frac{1}{2} m_1 R^2 \alpha}$$

$$2 R T_2 = \cancel{\left(\frac{1}{2} m_1 R^2 + m_1 R^2\right)} \alpha$$

$$\bar{M}^{e}_{polo} = I_z \alpha$$





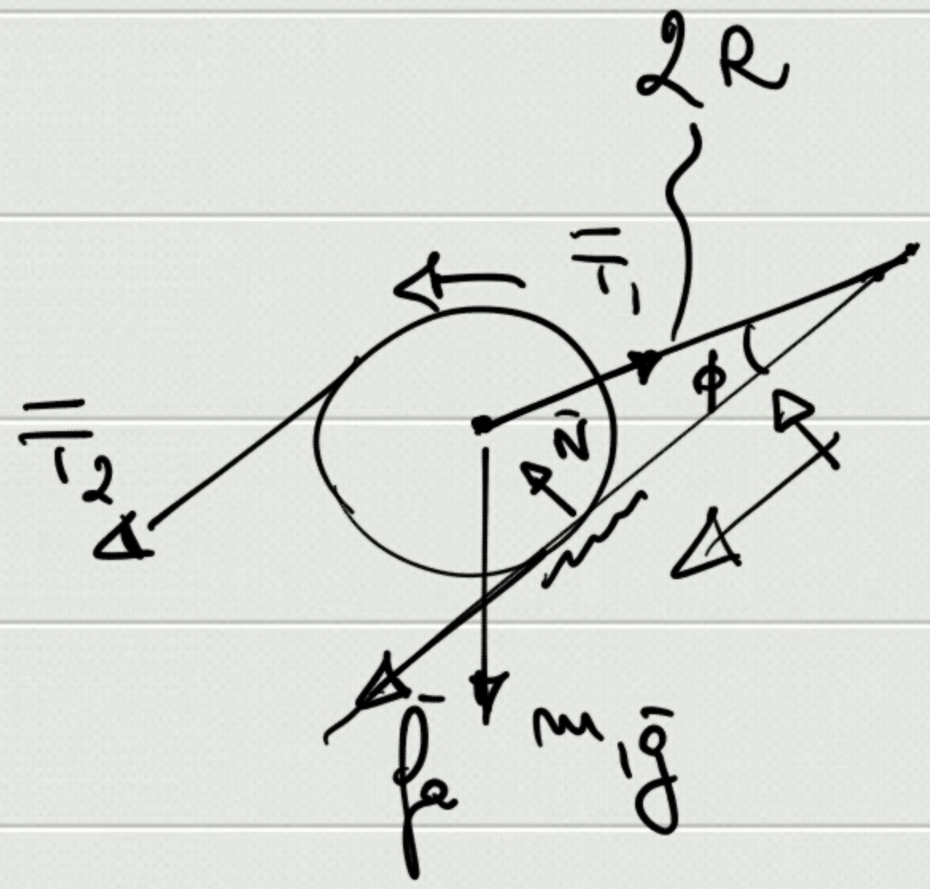


$$\begin{cases} m_2 g \sin\theta - T_2 = m_2 a_2 \\ \cancel{R} T_2 - \cancel{R} f_a = \frac{1}{2} m_1 R^2 \alpha = \frac{1}{2} m_1 R^{\cancel{2}} \frac{a_2}{\cancel{R}} \\ a_2 = a_T = \alpha R \end{cases}$$

$$m_2 g \sin\theta - f_a = \left(m_2 + \frac{1}{2} m_1\right) a_2$$

$$\Rightarrow a_2 = \frac{2\left(m_2 g \sin\theta - f_a\right)}{2m_2 + m_1} = 3 \ m/s^2$$

$$T_2 = f_a + \frac{1}{2} m_1 a_2 = 2.75 \ N$$

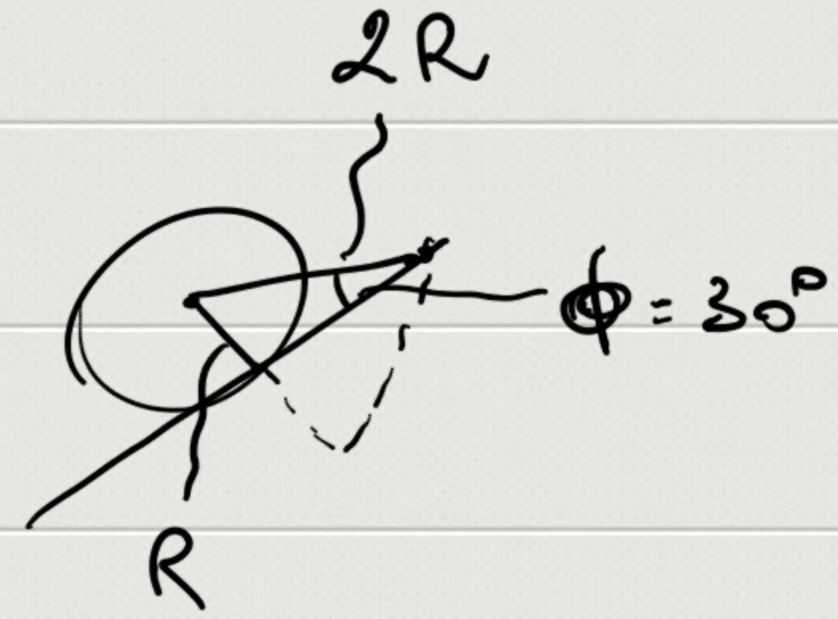


$T_1 = ?$

$$\bar{R}^{E} = m_1 \bar{a}_{CM} = 0$$

$$\boxed{\bar{T}_2 + \bar{f}_a + m_1\bar{g} + \bar{N} + \bar{T}_1 = 0}$$

$$\Rightarrow T_2 + f_a + m_1 g \sin\theta - T_1 \cos\phi = 0$$

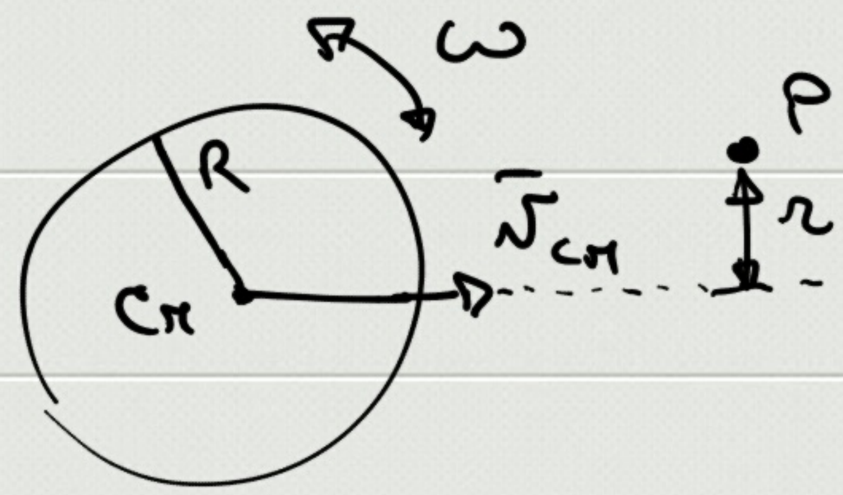


$$\Rightarrow T_1 = \frac{1}{\cos\phi}\left(T_2 + f_a + m_1 g \sin\theta\right) = 10.4 \text{ N}$$

$$f_a = \mu N \Rightarrow \mu = \frac{f_a}{N}$$

$$-m_1 g \cos\theta + N - T_1 \sin\phi = 0$$

$$\Rightarrow N = m_1 g \cos\theta + T_1 \sin\phi \qquad \Rightarrow \mu = \frac{f_a}{N} = 0.26$$



$R = 0.22\ m$

$v_{CM} = 3.6\ m/s$

$r = 0.16\ m$

$v' = \omega' = 0$

anello

$\bar{\omega} = ?$

$E_k = cost$ (urto elastico)

$\bar{P} = cost$

$\bar{P}(t_0^-) = m\bar{v}_{CM}$

$\bar{P}(t_0^+) = 0$

$\bar{J} = ?$ $\quad m = 1\,kg$

$\bar{J} = \Delta\bar{P} = \emptyset - m\bar{v}_{CM} = -m\bar{v}_{CM}$ $\quad \bar{J}$

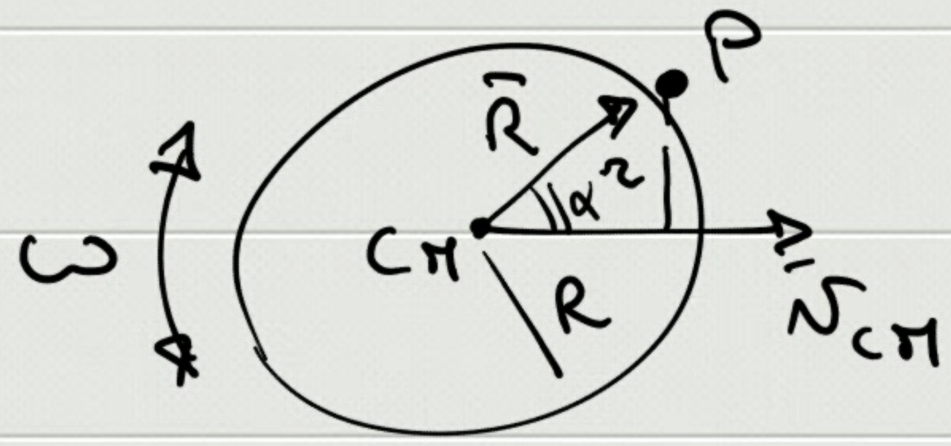
$J = m v_{CM} = 3.6\ Ns$

$\bar{L}_{polo} \overset{?}{=} \underline{\underline{cost}}$

$\bar{L}_P = cost$

$\overset{!}{=} \bar{L}_P(t_0^+) \overset{!}{=} 0$ $\quad \omega' = 0$

$\Rightarrow \boxed{\bar{L}_P(t_0^-) = 0}$



$$\bar{L}_P(t_0^-) = 0$$

$$0 = mR^2\bar{\omega} + \bar{R} \times m\bar{v}_{CM} \qquad *$$

$$0 = mR^2\bar{\omega} + m\bar{v}_{CM} \times \bar{R}$$

$$0 = -mR^2\omega + rmv \qquad * \qquad \otimes \bar{u}_z \qquad \odot \bar{\omega}$$

$$0 = (mR^2 + mR^2)\,\omega$$

$$\bar{L}_O = \underbrace{\bar{L}'_{CM}}_{\bar{I}_z\bar{\omega}} + \underbrace{\bar{L}_{O,CM}}_{\bar{r}_{CM} \times m\bar{v}_{CM}}$$

$$\bar{L}_O = \bar{I}_z\bar{\omega} \qquad \omega = \frac{rv}{R^2} = 11.9 \text{ rad/s}$$